Linux Fundamentals and Introduction to Networking Assignment

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Introduction

In this assigned TryHackMe rooms, I have understood the concept of Linux Fundamental and its common commands, how Network works, and Basic concept of LAN. TryHackMe is a free website which help Cybersecurity students, and persons who want to become ethical hackers to learn, and practices freely and easily. It incorporates different parts and essential points. TryHackMe provide easily navigable machines, resources, links, and strength the task gradually.

Linux provides the tools and operating system,Networking allows communication, and LAN(Local Area Network) connects devices locally.

1. Linux Fundamentals

TryHackMe | Linux Fundamentals Part 1

- Linux is the foundation, free, and open source Operation System.
- It provides tools, commands, and services to manage the system and networking.
- It works better than windows using more interacting approach.l
- I have also interacted with file system.
- Common examples of OS based on Linux
  - Ubuntu, and Kali Linux (Debian)
- The name "Linux " is an Umbrella term for multiple Operation System's which are based UNIX.

Remember:- Ubuntu server can run on systems with only 512MB of RAM.

In short, I have understood

- ✓ how I can solve flags using Linux commands, and run virtual machines, so it was really good as a beginner.
- ✓ I have watched each provide video which are helpful to understand the basic concept of each tryhackme rooms, and answer the give questions.
- ✓ In addition, I followed each instruction, hint, samples, and examples. As a result, I have solved each questions problems.
- ✓ Almost all Linux commands are case sensitive

Commands I practiced

"pwd" _____Print working directory which shows our current position.

"echo" _____ To write texts in the file

"whoami" _____To know the user name.

For example; my Linux terminal open as Solacyber@INSA, and want to know the user name, the command must be write like this.

Solacyber@INSA:-$whoami

Result:- Solacyber

"ls" _____ listing file, folders in the directory

"cd" _____ change directory, to list the full path of the current directory

"cat" _____ concatenate, to read the content of the file or message

find - name _____ to find a file with the name

find - name *.txt _____ if we don't know the file name.

grep _____ to search the entire contents of the file

grep -R _____ to search recursively through all files and sub directories

In addition there are also shell operator commands

"&" _____ to run commands on the background of our terminal

"&&" _____ to combine multiple commands together in one line of our terminal

">" _____ use to take the out put from a command, and direct it elsewhere.

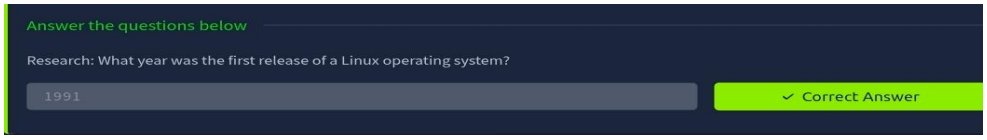
“>>” _____ almost the same with > command but it appends rather than replace.

In general, In this tryhackme room, I have understood how commands apply in practice to know my directory, search file, and folders, read, and write in the file. Plus to that Linux also has other commands which solve some limitation for example if searching a file using cat command is difficult we can use find, and grep command alternatively.

Challenges that I faced here are mixing up the commands application, and forgetting the case sensitivity of the Linux terminal.

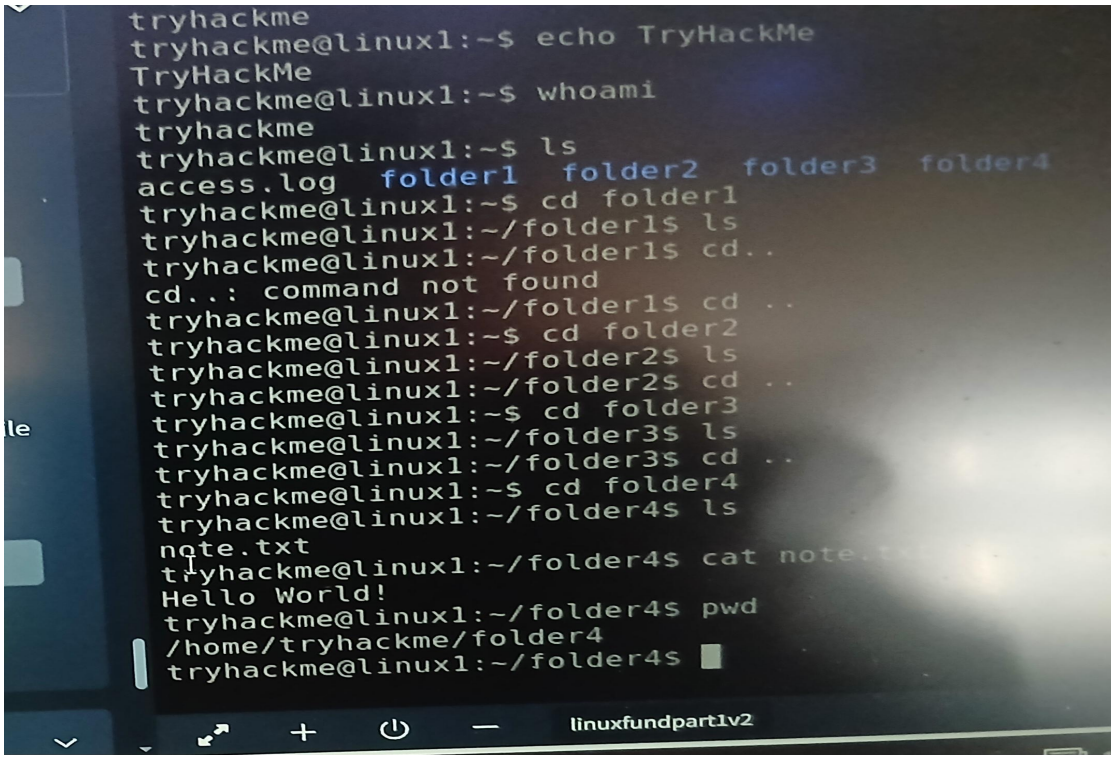
Answers for each Questions

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On The terminal

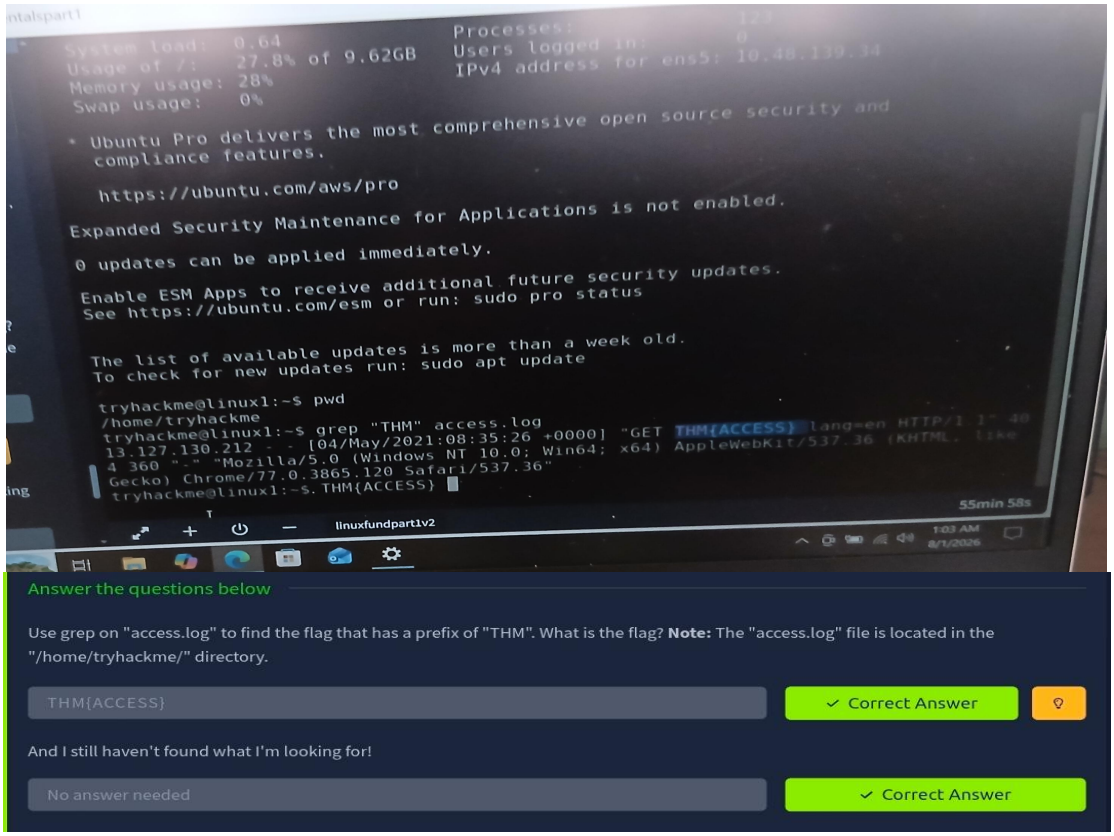




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On The terminal



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2. Networks

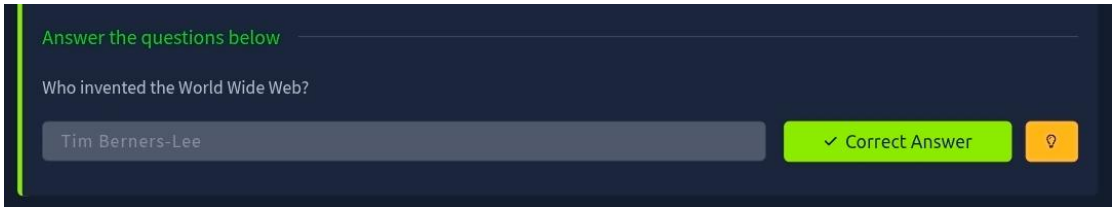
- This refers a connection or communication of devices to share files and resources.
- Network has its path to reach in the necessary or target place, and use some devices to facilitate that function like switch and router.
- The largest network in the world is **Internet**, commonly called “network of network”.
- IP(Internet protocol) Address that is the name of the network address. Devices use this name to communicate each other and deliver messages to the specific device. They divide into four octets, the first three parts are called network address, and the last octet is called host address. IP Address may be private, or public that helps our device to connect with the internet. IP Address can be change as we want
- MAC(Media Access Control) the unique address of our device, it is unchangeadle or fixed.
- IP Address can be divide into two based on version. IPv4 which is very common in this time., and IPv6 which is created to solve the limitation of IPv4, because it can cover around 2^{128} people in one network.
- Another point here is Ping(ICMP) is one of the most fundamental network tools available to us. It uses Internet Control Message Protocol(ICMP) packets to determine the performance of connection between devices. It explains detail epescilly how much time it takes and bytes.

Answer For Each Questions

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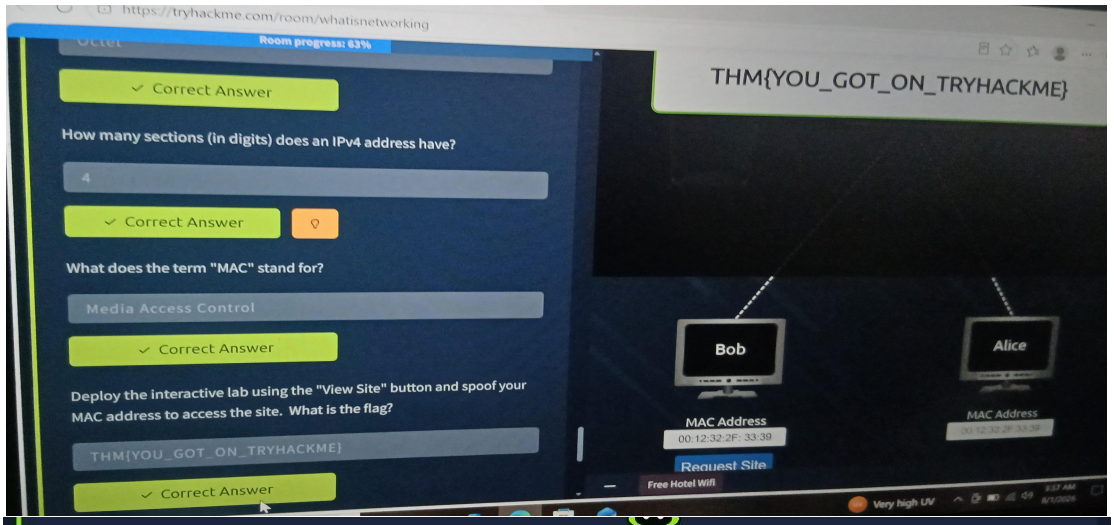


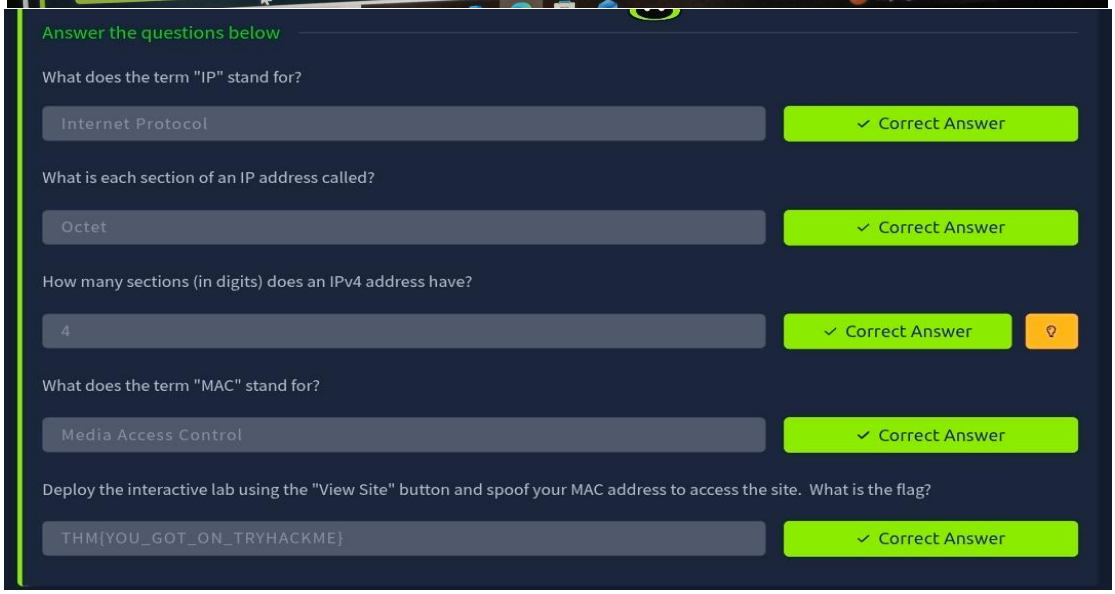
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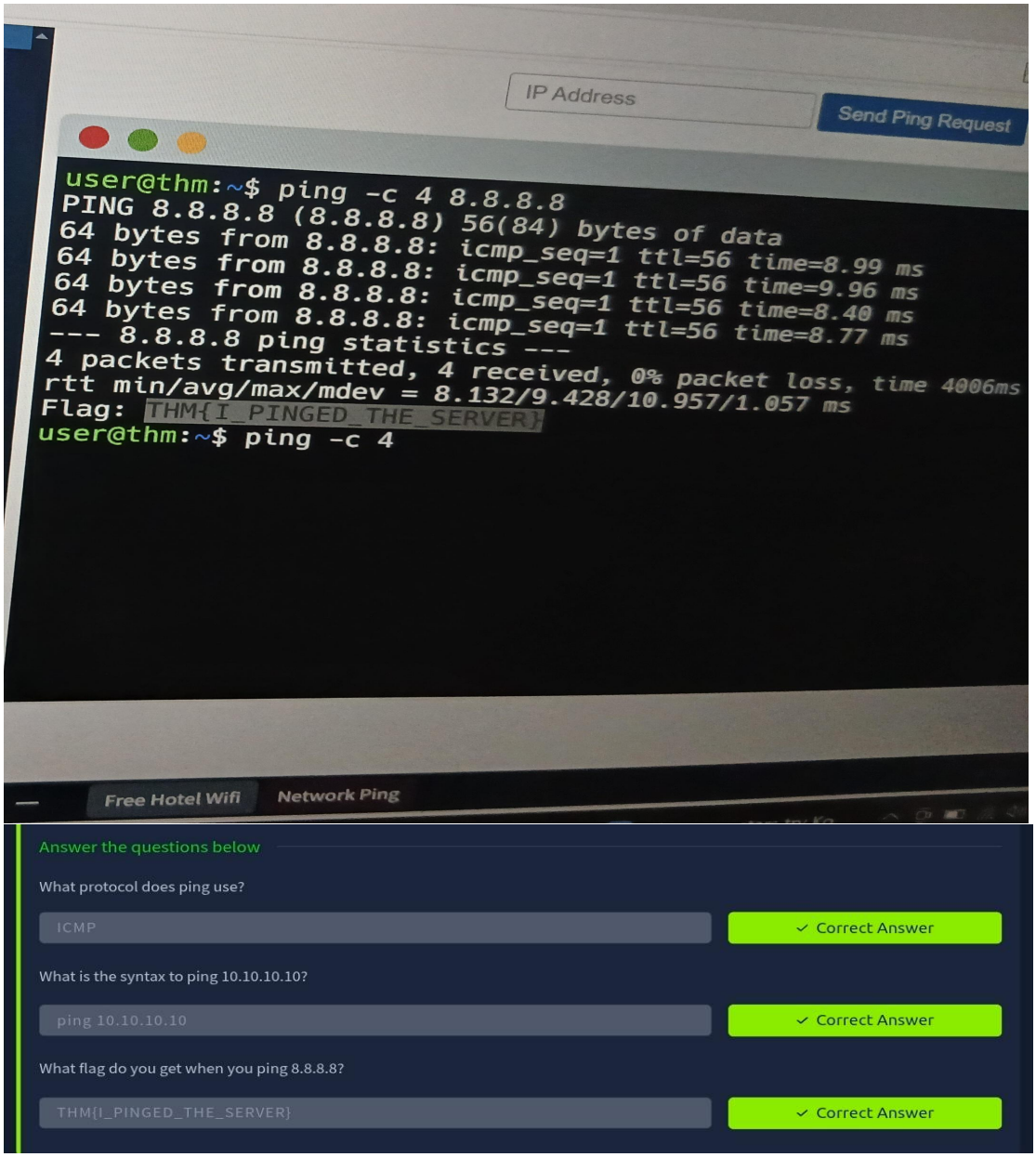
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On the terminal





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3. LAN (Local Area Networking)

- It is one of the networking types
- It covers small area lik home, school, office …etc.
- To answers the question how are devices connected in LAN network, use the term Topology which refers the physical or logical layout of how devices are connected in a networking.

  There are different types of Topology

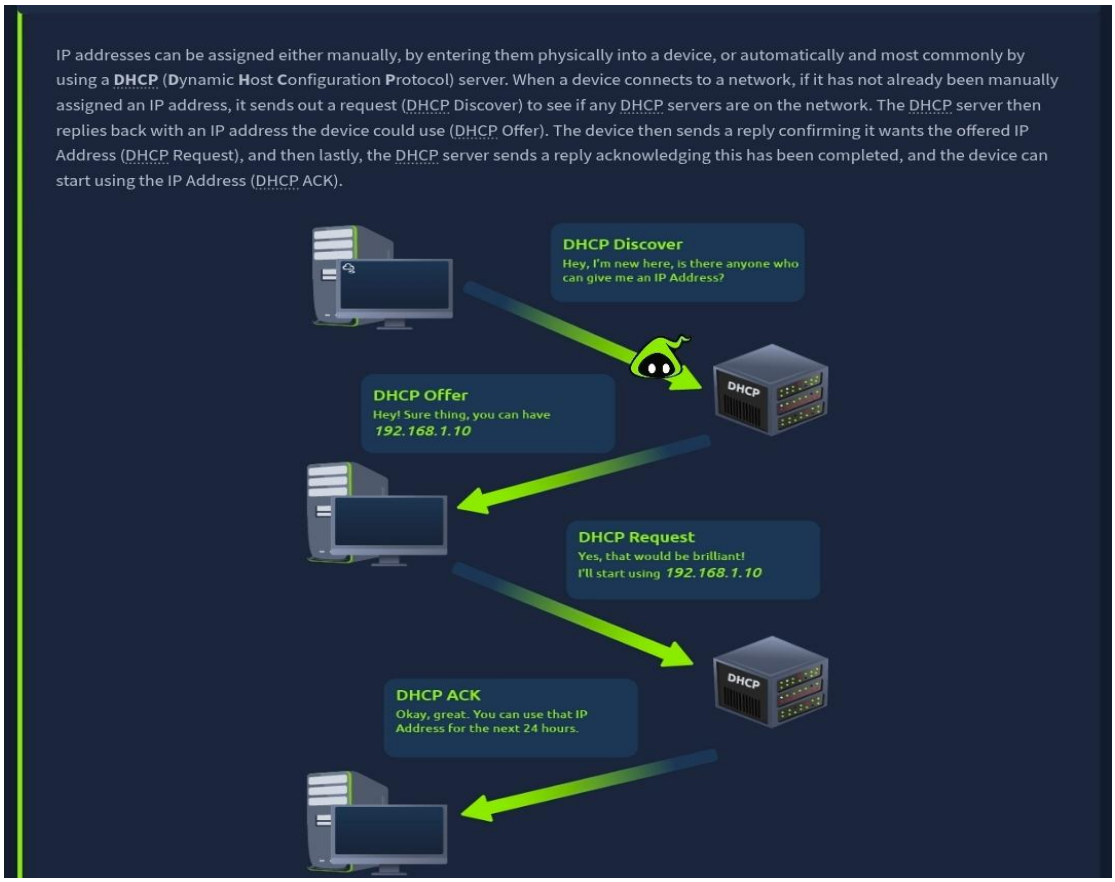
- ✧ Star topology:- every device connects to a central switch, but it is very expensive.
- ✧ Bus topology:- all devices connect to one main cable, and cost efficient.
- ✧ Ring topology:- devices form a circle, and it is ease to troubleshoot.

I have also covered about Subnetting, ARP, and DHCP

- Subnetting:- This a kind of study network address separately to know the specific address of the network like relative and absolute direction in geography. Subnets use IP address in three different ways:
  - The network Address
  - the host Address
  - the deafualt gate way

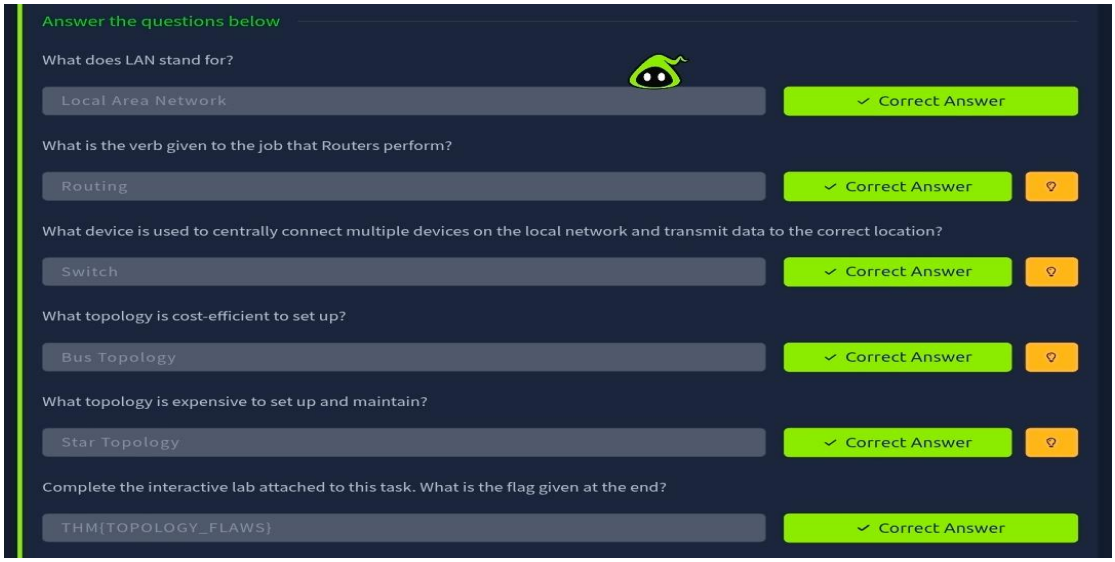
Subnet provide efficiency, security, and full control

- ARP(Address Control Resolution):- it help to change the IP address to MAC address, during this time ARP sends two message which are ARP request, and ARP reply.
- DHCP (Dynamic Host Configration Protocol);-

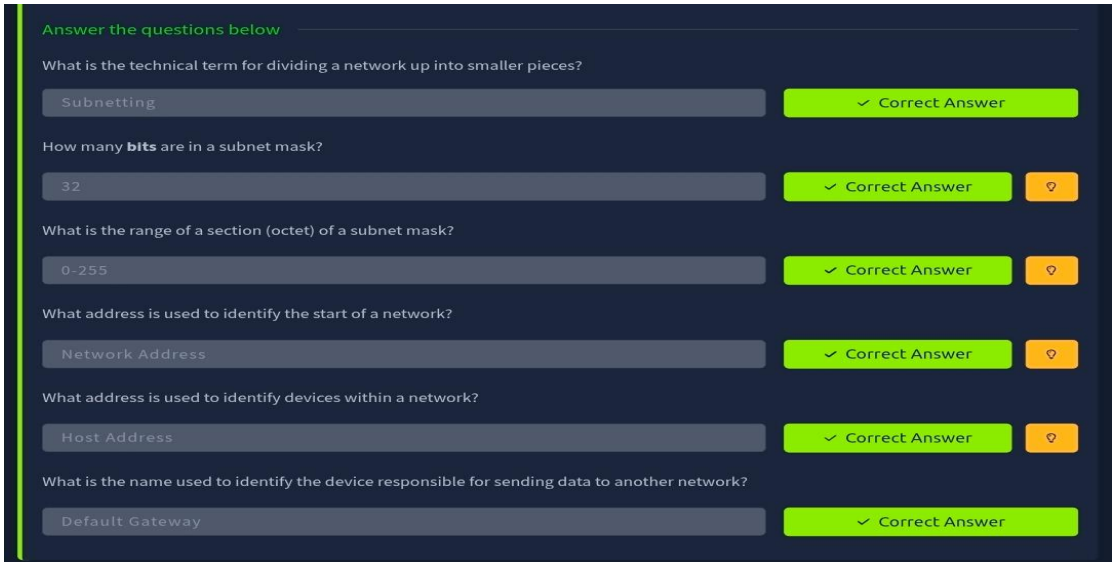


Answer For Each Questions

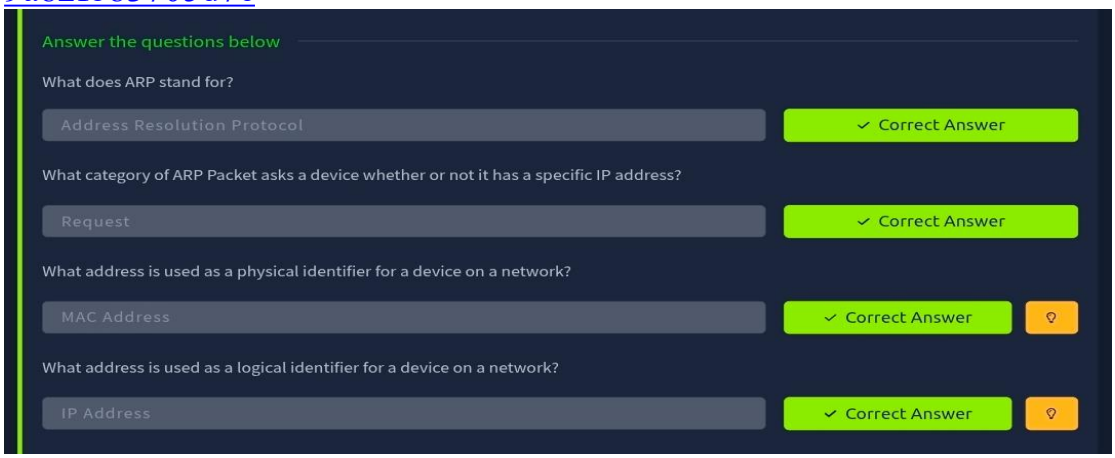
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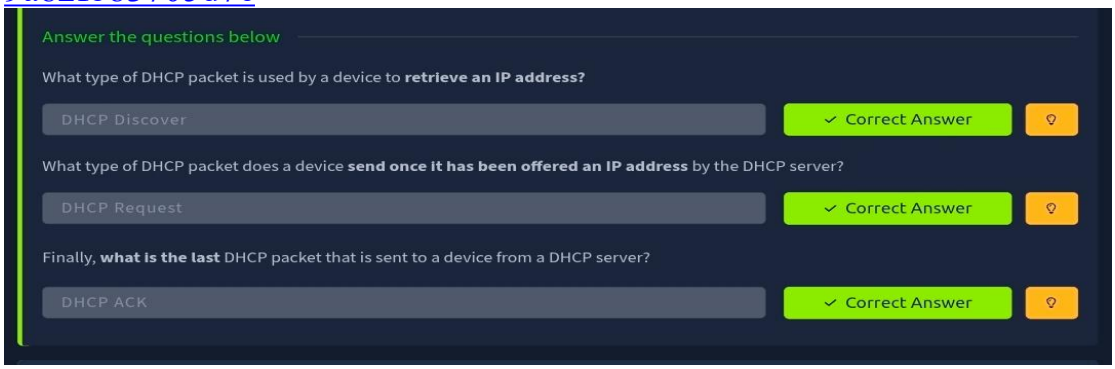
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Conculusion

In general, I have got the foundational concept about Linux, and network. I have understood about the Linux terminal, and practice by finding some CTF challenges. Moreover, I have seen how the network is working in this real world and ease the communication between devices.